

CLOQUET CARLTON COUNTY AP, MN, USA

WMO: 726558

Lat: 46.703N

Lon: 92.504W

Elev: 390

StdP: 96.73

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04914

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-24.5	-31.9	0.2	-27.2	-28.7	0.3	-23.3	10.6	-10.3	9.3	-11.7	3.2	290	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	29.0	21.3	27.6	19.9	26.2	18.8	23.3	27.1	21.8	25.8	20.5	24.3	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.8	26.1	20.9	16.3	24.4	19.0	14.4	22.7	71.5	27.3	65.7	26.3	60.9	24.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.4	7.5	-31.1	32.3	3.6	1.3	-33.7	33.3	-35.8	34.1	-37.8	34.9	-40.4	35.8		
			-31.2	25.2	3.5	1.4	-33.7	26.2	-35.7	27.0	-37.7	27.8	-40.2	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-11.6	-9.5	-2.9	3.9	10.3	15.5	19.1	18.0	13.7	6.7	-1.2	-8.4
	DBStd	11.90	7.45	7.08	6.70	4.83	4.30	3.80	3.32	3.20	4.42	4.64	5.96	6.66
	HDD10.0	2908	671	547	402	190	48	3	0	0	15	125	336	571
	HDD18.3	5152	929	780	659	433	251	100	31	44	149	361	584	829
	CDD10.0	916	0	0	1	7	58	168	282	249	126	23	1	0
	CDD18.3	117	0	0	0	0	2	15	55	35	10	0	0	0
	CDH23.3	1126	0	0	0	3	60	188	506	283	80	5	0	0
	CDH26.7	246	0	0	0	0	12	39	124	59	12	0	0	0
Wind	WSAvg	3.2	3.5	3.5	3.4	3.8	3.5	2.9	2.7	2.6	2.9	3.3	3.3	3.3
Precipitation	PrecAvg	786	28	21	41	59	85	106	103	101	98	68	48	29
	PrecMax	1080	106	60	109	202	204	215	196	232	270	201	195	72
	PrecMin	508	3	4	5	4	10	11	22	12	16	8	3	4
	PrecStd	133	21	14	24	38	37	47	44	55	54	49	37	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.9	17.1	22.4	28.0	30.9	32.2	31.2	27.9	23.0	16.2	6.2
		MCWB	2.5	4.2	11.1	11.7	17.1	23.3	23.2	22.8	20.5	14.9	10.6	2.6
	2%	DB	2.4	4.1	12.1	18.0	24.9	27.5	29.1	27.9	26.0	19.1	12.3	3.6
		MCWB	0.8	2.0	7.1	9.6	14.9	19.6	21.8	20.8	19.4	13.7	8.9	2.0
	5%	DB	0.1	2.4	8.0	15.9	22.2	25.9	27.7	26.4	23.0	17.1	8.9	2.1
		MCWB	-0.5	0.9	4.7	8.2	13.7	18.0	20.3	19.3	17.9	12.2	6.1	1.0
	10%	DB	-2.0	0.7	5.9	12.5	18.9	23.1	26.3	24.1	21.3	13.8	7.2	0.9
		MCWB	-2.6	-0.5	2.8	6.6	12.4	16.9	19.4	18.7	17.0	10.1	4.9	0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	5.0	11.9	14.1	19.2	24.1	25.2	24.8	22.9	17.7	12.4	3.8
		MCDB	5.3	7.1	16.3	18.3	23.7	28.3	29.5	28.6	26.7	19.8	14.0	4.8
	2%	WB	1.0	2.7	7.7	11.0	17.0	21.7	23.7	22.6	20.8	15.0	8.9	2.3
		MCDB	2.1	4.2	10.3	17.0	21.9	26.0	27.6	26.2	24.1	18.5	11.8	3.4
	5%	WB	-0.6	1.1	5.1	8.8	15.1	19.7	22.2	21.1	19.2	12.7	6.9	0.8
		MCDB	0.2	2.3	8.1	14.5	20.4	23.6	26.1	24.8	22.1	15.8	8.6	1.7
	10%	WB	-2.3	-0.9	3.0	7.1	13.2	18.0	20.7	19.8	17.6	10.9	4.9	-0.4
		MCDB	-2.0	0.2	5.3	11.5	18.0	21.3	24.7	23.3	20.1	13.5	6.9	0.5
Mean Daily Temperature Range		MDBR	9.0	9.8	9.8	10.7	12.1	11.8	11.5	10.9	10.5	8.9	7.7	7.5
	5% DB	MCDBR	8.8	10.0	12.3	16.9	17.4	15.2	13.2	12.9	13.1	12.9	11.0	8.5
		MCWBR	7.5	7.7	8.1	9.7	9.1	8.3	6.8	6.8	7.8	7.8	7.5	6.8
	5% WB	MCDBR	8.4	8.4	11.7	15.2	14.8	12.9	12.6	11.3	11.2	11.3	10.0	7.4
		MCWBR	7.3	6.7	8.1	9.5	9.1	8.6	7.7	7.1	7.5	7.8	7.5	6.3
Clear-Sky Solar Irradiance	taub	0.242	0.255	0.301	0.346	0.375	0.369	0.377	0.382	0.356	0.308	0.283	0.258	
	taud	2.400	2.374	2.377	2.375	2.343	2.384	2.385	2.391	2.438	2.566	2.524	2.423	
	Ebn at Noon	877	933	928	910	890	895	883	863	857	861	820	811	
	Edh at Noon	68	86	101	113	121	117	115	110	96	72	61	60	
All-Sky Solar Radiation	RadAvg	1.55	2.67	3.79	4.65	5.32	5.76	6.15	5.15	3.88	2.37	1.45	1.09	
	RadStd	0.17	0.23	0.36	0.46	0.40	0.40	0.25	0.39	0.29	0.32	0.15	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page